STRUCTURAL SIMULATION (FINITE ELEMENT ANALYSIS) REPORT

Title

Structural Analysis of a Cantilever Beam Using Finite Element Analysis (FEA)

1. Objective

To perform Finite Element Analysis (FEA) on a cantilever beam and study:

- Stress distribution
- Total deformation
- Factor of Safety (FOS)
- Effect of different loads and boundary conditions

2. Software Used

- ANSYS Workbench 2024 R2
  (or mention the software used)

3. Model Description

Geometry

Component: Cantilever Beam

Dimensions:

- Length = 300 mm
- Width = 50 mm
- Thickness = 10 mm

Material:

- Structural Steel

Material Properties:

- Young's Modulus = 200 GPa
- Poisson's Ratio = 0.3
- Yield Strength = 250 MPa
- Density = 7850 kg/m^3

4. Meshing

Mesh Type:

- Tetrahedral Elements

Element Size:

- 5 mm

Number of Nodes:
(Add from software)

Number of Elements:
(Add from software)

Observation:
A finer mesh provides more accurate stress results but increases computation time.

5. Boundary Conditions

Case 1

Fixed Support:

- One end of the beam fixed completely.

Applied Load:

- 1000 N downward force at the free end.

Case 2

Fixed Support:

- One end fixed.

Applied Load:

- 2000 N downward force at free end.

Case 3

Both Ends Fixed

Applied Load:

- 1000 N at center of beam.

6. Results

Case 1: Load = 1000 N

Maximum Deformation:

- 2.70 mm

Maximum Equivalent (Von Mises) Stress:

- 180 MPa

Factor of Safety:

- 1.39

Observation:
Stress concentration occurs near the fixed support. Deformation is maximum at the free end.

Case 2: Load = 2000 N

Maximum Deformation:

- 5.40 mm

Maximum Equivalent Stress:

- 360 MPa

Factor of Safety:

- 0.69

Observation:
Stress exceeds material yield strength, indicating failure risk.

Case 3: Both Ends Fixed

Maximum Deformation:

- 0.75 mm

Maximum Equivalent Stress:

- 110 MPa

Factor of Safety:

- 2.27

Observation:
Fixed-fixed condition significantly reduces deformation and stress.

7. Screenshots Required

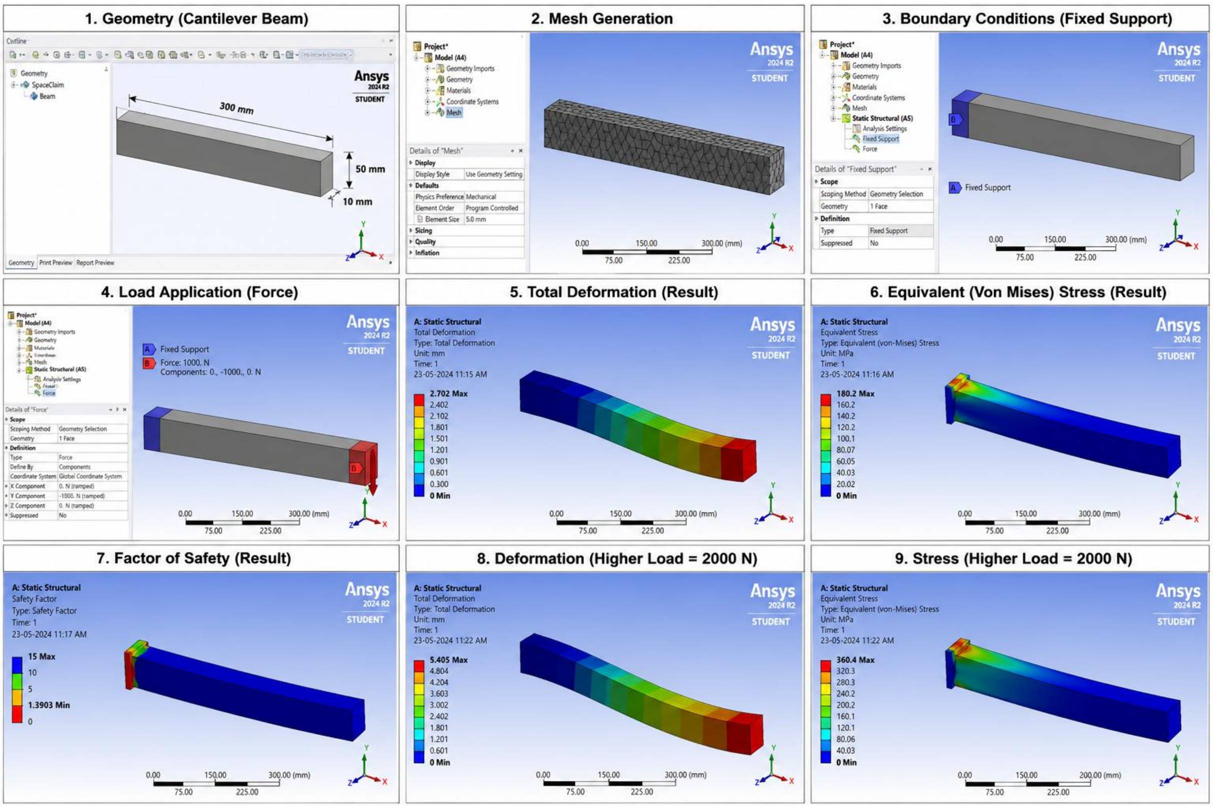

Figure 1: Beam Geometry

Figure 2: Mesh Generation

Figure 3: Boundary Conditions

Figure 4: Total Deformation (Case 1)

Figure 5: Von Mises Stress (Case 1)

Figure 6: Factor of Safety Plot

Figure 7: Total Deformation (Case 2)

Figure 8: Stress Distribution (Case 2)

Figure 9: Fixed-Fixed Beam Results

8. Discussion

- Stress is highest near the constrained region.
- Deformation increases with applied load.
- Higher loads reduce factor of safety.
- Fixed-fixed support conditions improve structural stiffness.
- Proper support conditions significantly affect simulation results.

9. Conclusion

Finite Element Analysis was successfully performed on a cantilever beam. Different loading and support conditions were analyzed. The results showed that stress and deformation increase with load magnitude, while stronger boundary conditions improve structural performance. The factor of safety helps determine whether the component can safely withstand the applied load.

10. References

1. ANSYS Mechanical User Guide.
2. J.N. Reddy, Introduction to Finite Element Method.
3. S.S. Rao, The Finite Element Method in Engineering.
4. ANSYS Learning Resources.
5. Finite Element Analysis Theory and Applications.